

The Role of Carbon Capture and Storage in Decarbonizing U.S. Data Centers

Hon Chung Lau* and Steve C. Tsai



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ABSTRACT: Analysis of public domain data shows that the power capacity of U.S. data centers (DCs) will increase from 40 GW in 2025 to 169 GW in 2030, a 4.2-fold increase. The states with the highest growth rate in DC power capacity are Texas, Virginia, Pennsylvania, Ohio, Arizona, Colorado, Utah, and Illinois. This study uses the power capacity and announced energy source of DCs as well as the energy mix of the state grid to estimate the CO₂ emitted from fossil-power plants supplying electricity to DCs (DC-emitted CO₂). DC-emitted CO₂ was estimated to be 90 Mtpa in 2025. This number will grow to 404 Mtpa in 2030, a 4.5-fold increase. Among subsurface reservoirs in the U.S., over 95% of CO₂ storage capacity resides in saline aquifers, which have a total CO₂ storage capacity of 11.37×10^{12} tons. These saline aquifers can be used for sequestering CO₂ emitted from both gas-fired (GFPPs) and coal-fired power plants (CFPPs). Results of a CO₂ source-sink mapping exercise show that 34 states have enough storage capacity in saline aquifers for over 100 years of CO₂ sequestration, beyond 2030. Saline aquifers in the U.S. can sequester 59 Mt or 66% of DC-emitted CO₂ in 2025. This number will grow to 299 Mt or 74% in 2030. Of the 16 states that do not have adequate CO₂ storage capacity, six have the option of sequestering CO₂ in a bordering state. These results suggest that carbon capture and storage (CCS)-equipped fossil fuel power plants are a viable option to provide low-carbon electricity to U.S. DCs. Furthermore, natural gas combined cycle (NGCC) power plants equipped with CCS are preferred over CFPPs due to the abundance of gas reserves, the proximity of gas fields to saline aquifers, and the lower CO₂ emission of GFPPs compared to CFPPs.



1. INTRODUCTION

DCs house the servers necessary for digital infrastructure, cloud computing, artificial intelligence (AI), machine learning, e-commerce, and data storage. By centralizing the processing power required for these computations, DCs enable businesses to scale rapidly, innovate through real-time data analytics, and maintain the seamless connectivity that defines modern day commerce.^{1,2} In essence, DCs transform raw data into economic utility with logistical efficiency and drive economic growth through massive capital investments, the creation of high-paying jobs, and tax contributions to governments.³

Over the past decade, DCs have experienced explosive growth.⁴ In particular, their growth accelerated significantly after 2017, with hyperscale capacity doubling between 2020 and 2025.^{5,6} Supporting this rapid growth in DCs requires enormous physical and digital infrastructure, with power availability being one of the top criteria for site selection. Besides power, DC growth requires advanced, high-density cooling technologies, such as liquid and immersion cooling, to manage the intense heat generated by AI hardware. Additionally, DCs require specialized skilled labor, optimizing supply chains for critical components such as generators and transformers and securing sites with high-capacity fiber connectivity. Finally, sustainable DC growth depends on

reducing the CO₂ footprint through the use of low-carbon electricity and implementing best practices of a circular economy to minimize environmental impact and manage e-waste.⁷

There have been various studies on the power requirement of DCs in the U.S.⁸ These studies use one of three approaches: bottom-up, top-down, or extrapolation. The bottom-up approach is based on equipment specification, such as server power requirement and power usage effectiveness.^{9–16} The top-down approach is based on government or organizational measurement, such as total data volume and total operator power consumption.¹⁷ The extrapolation approach uses a baseline value from either the bottom-up or top-down model and assumes a growth rate by factoring in energy efficiency improvements and growth in data volumes.^{18,19} Of the three, the bottom-up approach is recognized for its robustness due to

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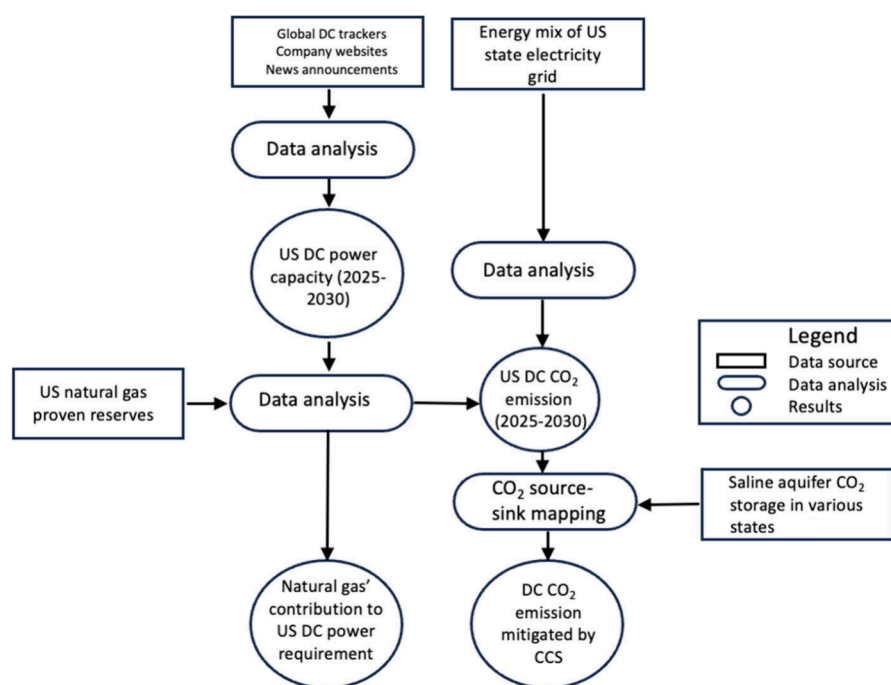


Figure 1. Methodology of study.

its reliance on meticulous data collection of information technology (IT) equipment inventory within DCs. However, all three approaches suffer from the unavailability of organizational databases and inability to project future trends over more than several years.

A recent report by the U.S. Department of Energy (DOE), based on the bottom-up approach, finds that DCs consumed about 4.4% of total U.S. electricity in 2023 and that this number will increase to 6.7–12% by 2028.¹⁶ The report indicates that total DC electricity usage climbed from 58 TWh in 2014 to 176 TWh in 2023 and estimates an increase to between 325 and 580 TWh by 2028. However, because of its bottom-up approach, the projected DC growth rate in power consumption, though significant, may still be too conservative.

This study focuses on the decarbonization of U.S. DCs. Its novelty consists of the following. First, unlike most studies on DCs, it focuses not on power consumption but rather on the power capacity of DCs. This is because DCs are designed according to power capacity rather than electricity consumption.^{20–22} Once a facility is built, the addition of capacity will require major renovation, whereas power consumption can vary within a certain range. In estimating the power capacity of DCs, we use a top-down approach by considering the DC power requirement from all available sources in the public domain. Second, this study calculates the DC-emitted CO₂, which is often neglected in other studies. Third, this study determines how much of the DC-emitted CO₂ can be mitigated by CCS²³ since it is recognized as one of the most mature technologies in large-scale decarbonization.^{24–36} This is achieved by conducting a state-by-state CO₂ source-sink analysis.

2. OBJECTIVES AND METHODOLOGY

There are three objectives for this study. First, we estimate the power capacity of U.S. DCs in the six year period between 2025 and 2030 based on public domain data. Second, we estimate the CO₂ emission from U.S. DCs in this six year period based on the fraction of DCs

powered by fossil fuels. The results are used in a CO₂ source-sink mapping exercise to determine how much of the DC-emitted CO₂ can be sequestered in saline aquifers. From this, we estimate the contribution of CCS in decarbonizing U.S. DCs. Third, we estimate the contribution of natural gas in supplying power for future U.S. DCs based on projected power generation requirements and the known proven natural gas reserves in the U.S. The results of this study can be used in several ways. First, they allow the power industry to determine the required addition to their current capacity to power future DCs in various states. Second, they provide the data for DC owners to consider using natural gas with CCS as a low-carbon energy source for future DCs. Third, they can be used by policymakers to promulgate energy policies related to DCs.

Figure 1 illustrates the methodology used in this study. Information on U.S. DCs is obtained from three major global DC trackers,^{37–39} company Web sites,^{40–44} and news announcements.^{45–59} We include only DCs where the required power capacity is announced. Next, we considered the power source of these DCs. If the power source is not announced, we assume it is provided by the state grid. Then, we use the energy mix of the state grids to determine the CO₂ emission from the DCs.⁶⁰ If the announced power source is nuclear or renewable energies, e.g., hydroelectricity, solar photovoltaic (PV), or wind, we assume the power plant will not emit CO₂ during its operation. This approach is conservative because it excludes future DCs where the power capacity is not disclosed for competitive or other reasons.

The three major types of subsurface reservoirs where CO₂ can be stored are oil reservoirs, gas reservoirs, and saline aquifers.²³ Unminable coal can be used for CCS but is not considered in this study since this technology is less developed. According to estimates by the DOE, over 95% of the storage capacity is found in saline aquifers.⁶¹ In this study, we use the CO₂ storage capacity of saline aquifers published in the *Carbon Storage Atlas* of the DOE.⁶¹ We perform a CO₂ source-sink mapping exercise to determine the amount of DC-emitted CO₂ that can be sequestered in in-state saline aquifers. In this study, a state is deemed to have an adequate CO₂ storage capacity if its saline aquifers can sequester more than 100 years of CO₂ injection at the 2030 injection rate. If this criterion is not satisfied, then we consider the option of transporting CO₂ to a bordering state for sequestration.

Next, we estimate the cumulative power consumption by U.S. DCs between 2025 and 2030 as well as the volume of natural gas needed to

provide this power by using typical energy efficiencies of gas turbines. We then compare that with the proven natural gas reserves in the U.S. to determine whether there are enough natural gas reserves to power the DCs. Finally, we compare the levelized cost of electricity (LCOE) for various forms of low-carbon electricity to power DCs in the U.S.

3. RESULTS

3.1. Power Capacity Demand by U.S. DCs

Table 1 and Figure 2 show the power capacity of DCs in all 50 U.S. states from 2025 to 2030 based on an analysis of all

Table 1. U.S. DC Power Capacity in 2025 and Estimated Growth up to 2030

State	DC power capacity in 2025 (GW)	Estimated DC power capacity in 2030 (GW)	Increase in DC power capacity between 2025 and 2030 (GW)
Texas	6.847	31.969	25.122
Virginia	8.322	24.419	16.097
Pennsylvania	1.908	12.752	10.844
Ohio	0.892	9.726	8.834
Arizona	2.002	8.945	6.943
Colorado	0.478	7.478	7.000
Utah	0.222	7.212	6.990
Illinois	1.746	7.079	5.333
Indiana	0.061	4.861	4.800
Georgia	1.595	4.782	3.187
Nevada	0.777	4.625	3.848
Louisiana	2.021	4.321	2.300
North Carolina	1.016	4.152	3.136
Mississippi	0.323	3.667	3.344
California	1.989	3.840	0.851
Iowa	0.660	2.600	1.940
North Dakota	1.090	2.422	1.332
Delaware	0.008	2.408	2.400
Tennessee	0.338	2.058	1.720
West Virginia	0.335	1.991	1.656
Washington	1.705	1.840	0.135
Maryland	0.672	1.820	1.148
New Jersey	0.961	1.575	0.614
Michigan	0.127	1.531	1.404
Wyoming	0.204	1.439	1.235
New Mexico	0.058	1.208	1.150
Oregon	0.732	1.142	0.410
Connecticut	0.791	1.015	0.224
Others	2.187	7.067	4.880
Total	40.067	168.943	128.876

available public domain information. The states with the highest DC power capacity in 2030 will be Texas, Virginia, Pennsylvania, Ohio, and Arizona (Figure 2). Texas, Virginia, and Pennsylvania have to increase their power capacity by 25, 16, and 11 GW, respectively, to provide enough power to their DCs. Between 2025 and 2030, Ohio, Arizona, and Colorado have to add 8.8, 5.9, and 7.0 GW, respectively, to satisfy the power demand of their DCs. Part of the reason for this increase is the rapid growth of AI campuses with power and cooling requirements from next-generation semiconductor chips.

The total DC power capacity of all 50 states will increase from 40 GW in 2025 to 169 GW in 2030, a 4.2-fold increase (Figure 3), corresponding to a compound annual growth rate (CAGR) of 43.3%. However, the CAGR of total U.S. power capacity between 2025 and 2030 is expected to be only 2.7%.⁶²

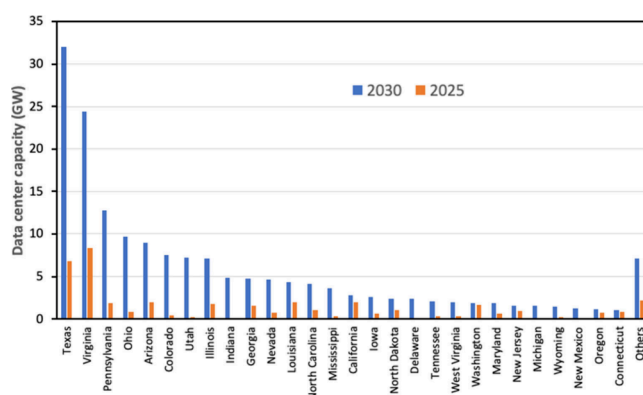


Figure 2. Increase in DC power capacity from 2025 to 2030 in various U.S. states.

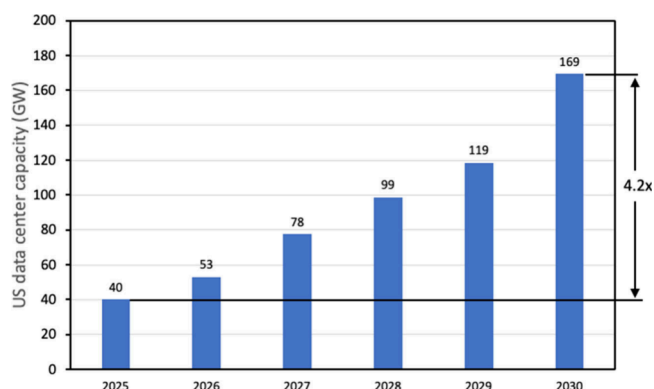


Figure 3. Estimated U.S. DC power capacity between 2025 and 2030.

The total U.S. DC power capacity will increase from 3.3% of the U.S. total in 2025 to 12.3% in 2030 (Figure 4). In 2030, 18

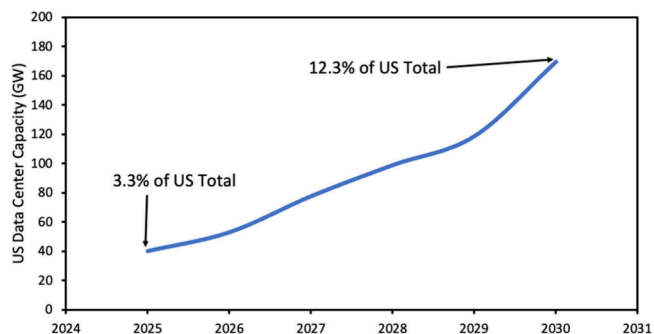


Figure 4. Total U.S. DC power capacity from 2025 to 2030.

states will have a DC power capacity exceeding 10% of the total state power capacity (Figure 5), thus requiring the addition of new capacity. They are Virginia, Delaware, Utah, Colorado, Ohio, Nevada, Arizona, Pennsylvania, North Dakota, Mississippi, Texas, Louisiana, Indiana, Maryland, Illinois, Wyoming, West Virginia, and Georgia.

Figures 6 and 7 show the location of DCs and their power capacity in the lower 48 states in 2025 and 2030, respectively. DCs are clustered in certain areas due to reasons such as proximity to cable network hubs and customers, cost of electricity, location security, and avoidance of natural disasters. In 2025, states with the biggest clusters of DCs are Virginia, Texas, Arizona, Louisiana, and California (Figure 6). By 2030,

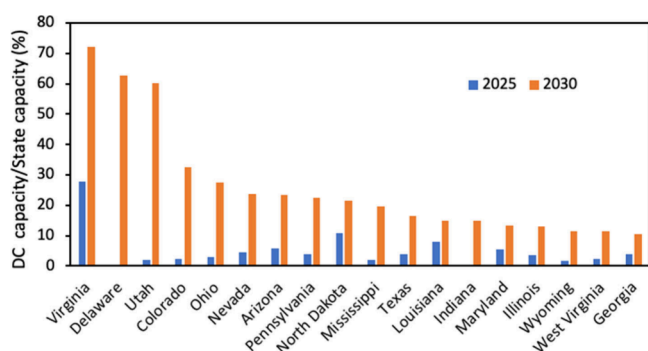


Figure 5. States where DC power capacity in 2030 will exceed 10% of the total state power capacity.

big DC clusters also appear in Ohio, Colorado, Utah, Illinois, Indiana, Georgia, Nevada, North Carolina, Mississippi, Iowa, North Dakota, Delaware, and Tennessee (Figure 7). The top 10 states with the biggest increase in DC power capacity between 2025 and 2030 are Texas, Virginia, Pennsylvania, Ohio, Arizona, Colorado, Utah, Illinois, Indiana, and Nevada (Table 1). Interestingly, there is little increase in DC power capacity in California, probably because of its very high cost of electricity.

3.2. Power Supply for U.S. DCs

In 2023, power generation in the U.S. was 43% by natural gas, 19% by nuclear, 16% by coal, 15% by nonhydro renewables, and 6% by hydroelectricity (Figure 8), based on analysis of data published by the U.S. Energy Information Administration (EIA).⁶³ Data centers have a very high uptime per year requirement, ranging from 99.67% for small businesses (Tier I) to 99.995% (Tier IV) for large enterprises with international reach.⁶⁴ There is no power redundancy requirement for Tier I DCs, but full redundancy is required for Tier IV DCs to avoid the permanent loss of data and costly contractual penalties. This suggests that DCs should be powered by power plants with the highest reliability and uptime performance. One way to measure this is the capacity factor of a power plant.^{65–67} It is

defined as the ratio of actual electricity produced by a plant over one year to the maximum electricity that can be generated if the plant is operated at full capacity continuously. Table 2 compares the capacity factor of various energy sources in the U.S. in 2024.⁶⁵ Nuclear, natural gas combined cycle, and coal have capacity factors of 92%, 60%, and 43%, respectively. Hydroelectricity, wind, and solar PV have capacity factors of 35%, 35%, and 23%, respectively. The low capacity of wind and solar PV is due to their intermittency, as they cannot provide electricity on a 24/7 basis. Therefore, neither one can provide the base-load for a power plant, unless expensive energy storage is installed. Licensing of nuclear power plants usually requires many years from proposal to actual completion.⁶⁸ Even small modular reactors (SMRs) will not be fully commercial before 2030.⁶⁹ At present, NGCC is the most reliable source of power supply for future DCs based on mature technology.^{70–73}

3.3. CO₂ Emission from U.S. DCs

Next, we calculate the CO₂ emission from DCs that are powered by fossil fuel power plants connected to the state grids.⁶⁷ We use the state electricity energy mix, given in Table S1 in the Supporting Information, and the power capacity of DCs in each state to determine the power capacity of three types of fossil power plants for DCs: CFPPs, (GFPPs), and oil-fired power plants (OFPPs). We then multiply these power capacities by conversion factors, given in Table S2, to determine the yearly CO₂ emission. The conversion factor for CFPPs is obtained from EIA.⁷⁴ Conversion factors for OFPPs and GFPPs are obtained from IPCC⁷⁵ with supporting data from IEA^{76,77}. We further assume that the state energy mix will remain the same until 2030 to calculate the DC-emitted CO₂ in 2030. The results are shown in Table 3. In 2025, the DC-emitted CO₂ was 90 Mt, which was approximately 1.8% of the total U.S. CO₂ emission. In 2030, this number will grow to 404 Mt, a 4.5-fold increase (Figure 9), with a net increase of 315 Mt (Table 3). Figure 10 shows the increase in DC-emitted CO₂ from 2025 to 2030 for various U.S. states. The top 10 states in this regard are Texas, Virginia,



Figure 6. DC clustered areas and power capacity in the lower 48 states of the U.S. in 2025.

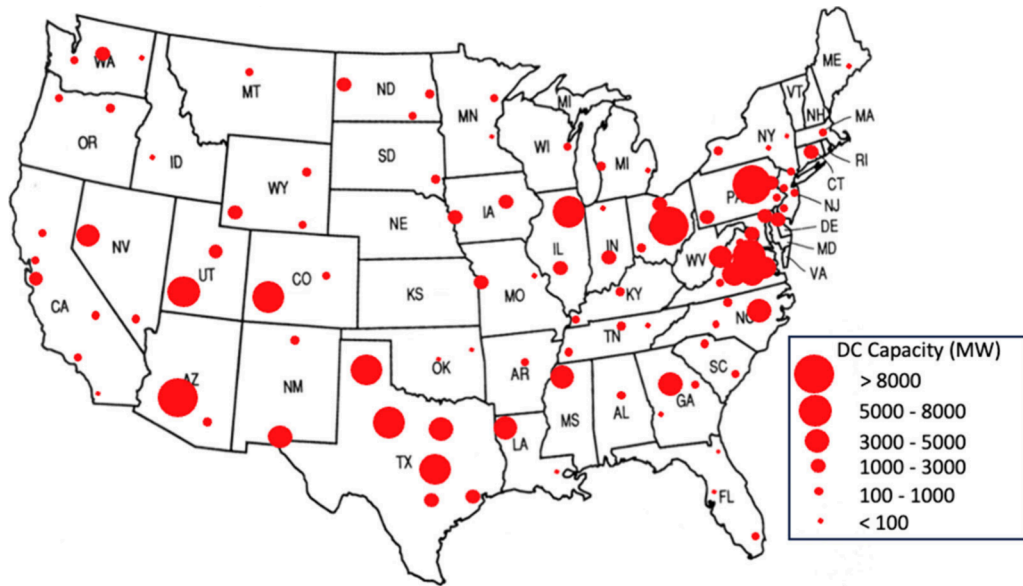


Figure 7. Estimated DC clustered areas and their power capacity in the lower 48 states of the U.S. in 2030.

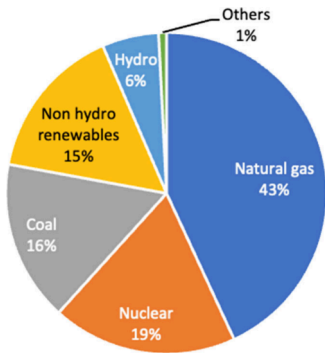


Figure 8. Power generation in the U.S. by fuel source in 2023 (authors' results).

Table 2. Capacity Factor for Selected Energy Sources in the U.S. in 2024 (Source: Data from EIA (2026)⁶⁷ CC BY 4.0)

Power source	Capacity factor (%)
Nuclear	92.3
Natural gas combined cycle	59.7
Coal	42.6
Hydroelectricity	34.5
Wind	34.5
Solar photovoltaic	23.4

Ohio, Colorado, Illinois, Indiana, Pennsylvania, Arizona, North Carolina, and Utah.

Figures 11 and 12 show the magnitude of CO₂ emission from the lower 48 states in 2025 and 2030, respectively. It can be seen that there is a significant increase in DC-emitted CO₂ in this six year period. Table 4 shows the U.S. cities where DC-emitted CO₂ will be over 5 Mtpa in 2030. There will be 14 cities where DC-emitted CO₂ will be over 10 Mtpa and nine cities with DC-emitted CO₂ between 5 and 10 Mtpa. In 2025, there was no cluster with DC-emitted CO₂ exceeding 10 Mtpa. In 2025, the top five cities with the biggest DC-emitted CO₂ were Ashburn, Chicago, Shreveport, Dallas-Fort Worth, and Austin. In 2030, the top five cities will be Columbus, Delta, Chicago, Austin, and Phoenix.

Table 3. Estimated CO₂ Emission of U.S. DCs in 2025 and 2030

State	DC-emitted CO ₂ in 2025 (Mtpa)	DC-emitted CO ₂ in 2030 (Mtpa)	Increase in DC-emitted CO ₂ from 2025 to 2030 (Mtpa)
Texas	16.49	73.57	57.08
Virginia	18.69	58.21	39.53
Ohio	2.93	32.97	30.04
Colorado	1.18	23.97	22.79
Illinois	6.57	23.90	17.33
Indiana	0.23	18.30	18.07
Arizona	3.89	17.39	13.49
Pennsylvania	1.74	17.29	15.54
Louisiana	6.33	14.98	8.65
Mississippi	1.21	12.40	11.19
North Carolina	2.01	11.50	9.49
Utah	0.77	10.74	9.98
Georgia	3.47	10.51	7.03
Nevada	1.65	9.72	8.06
Delaware	0.03	8.53	8.50
North Dakota	3.34	7.43	4.09
West Virginia	0	6.23	6.23
Maryland	1.83	5.44	3.61
Tennessee	1.03	5.16	4.14
Iowa	1.43	4.30	2.87
New Jersey	1.63	30.8	1.45
Kansas	0.59	3.02	2.43
Missouri	0.59	3.02	2.43
California	2.03	2.90	0.87
Kentucky	0.96	2.89	1.93
New Mexico	0.14	2.46	2.32
Connecticut	1.65	2.12	0.47
New York	0.53	1.54	1.01
Michigan	0.35	1.47	1.12
Washington	1.34	1.44	0.10
Others	5.03	7.76	2.74
Total	89.67	404.25	314.58

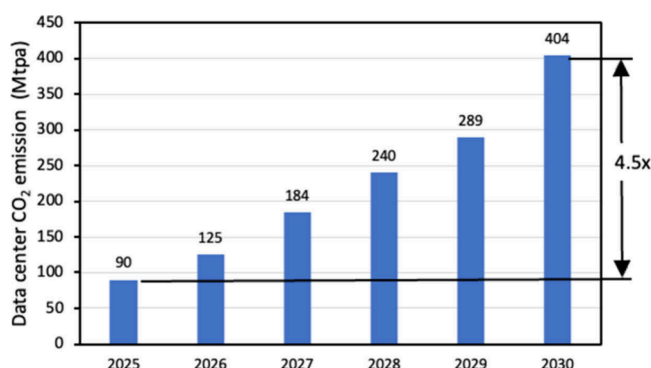


Figure 9. Estimated CO₂ emission from U.S. DCs between 2025 and 2030.

3.4. Contribution of CCS in Decarbonizing Fossil Fuel-Powered DCs in the U.S.

There are three major types of subsurface reservoirs for the storage of CO₂: oil reservoirs, gas reservoirs, and saline aquifers.²³ The DOE has calculated the subsurface CO₂ storage capacity in these reservoirs and found that over 95% of the storage capacity resides in saline aquifers.^{23,61} Therefore, in this study, we consider using only saline aquifers for CO₂ sequestration. Figure 13 shows the location of saline aquifers relative to locations of DC-related CO₂ emission. It can be seen that many emission locations reside within or near a saline aquifer.

3.4.1. In-State CCS Option. Table 5 gives the CO₂ storage capacity of saline aquifers and DC-emitted CO₂ in 2025 and 2030 for all 50 states in the U.S. The DC-emitted CO₂ in the U.S. was 90 Mtpa in 2025. In comparison, the CO₂ emitted from the U.S. cement,⁷⁸ steel,⁷⁹ and transportation⁸⁰ industries were 71, 95, and 1870 Mtpa, respectively.

The CO₂ storage capacity in saline aquifers can be calculated by using the following formula:⁶¹

$$m_{\text{CO}_2} = \rho_{\text{CO}_2} \times A \times h \times \phi \times E \quad (1)$$

where m_{CO_2} is the mass of CO₂ stored, ρ_{CO_2} is the density of CO₂ at reservoir temperature and pressure, A is the aquifer area in m², h is the net sand thickness in m, ϕ is the formation porosity (fraction), and E is the CO₂ storage efficiency factor, which reflects the fraction of the total pore volume of the saline aquifer that will be occupied by the injected CO₂. It should be noted that uncertainty in the variables A , h , ϕ , and E will result in uncertainty in m_{CO_2} . In this study, we use the medium m_{CO_2} estimated by the DOE⁶¹ (Figure 14). The low and high values can be considerably different from the medium value.

The total medium CO₂ storage capacity is 11.37×10^{12} tons. The number of years of injection is obtained as follows. We assume that CO₂ injection started in 2025 and continued to 2050 and that the totality of DC-emitted CO₂ is injected into saline aquifers. We then divide the remaining CO₂ storage capacity by the CO₂ emission rate in 2030. Our calculations show that in 34 states, the saline aquifers have enough remaining capacity to store more than 100 years of injection beyond 2030. In these 34 states, 59.1 Mtpa of DC-emitted CO₂ can be mitigated by CCS. This is about 1.2% of the total U.S. CO₂ emission. By 2030 or after 6 years of CO₂ injection, the amount of DC-emitted CO₂ that can be mitigated by CCS increases to 299.2 Mtpa, representing 73.5% of the total DC-emitted CO₂. This is estimated to be 8.1% of the total U.S. CO₂ emission. This result reveals the suitability of CCS in decarbonizing fossil fuel-powered DCs.

Among the 38 states with available CCS storage capacity, four states will run out of capacity (<100 years of injection) by 2030 or within 6 years from the start of CO₂ injection in 2025. Two of the four states are Virginia and Arizona, which are among the top 10 states with DC-emitted CO₂.

Furthermore, 12 states have no saline aquifers for CO₂ sequestration. However, these states have little DC-emitted CO₂ except Nevada, Iowa, New Jersey, and Connecticut. These four states may consider out-of-state CCS options. The offshore CCS option is not considered in this study.

3.4.2. Out-of-State CCS Option. In the eight states with no or inadequate in-state CO₂ storage capacity in saline aquifers, sequestration in a bordering state may be considered

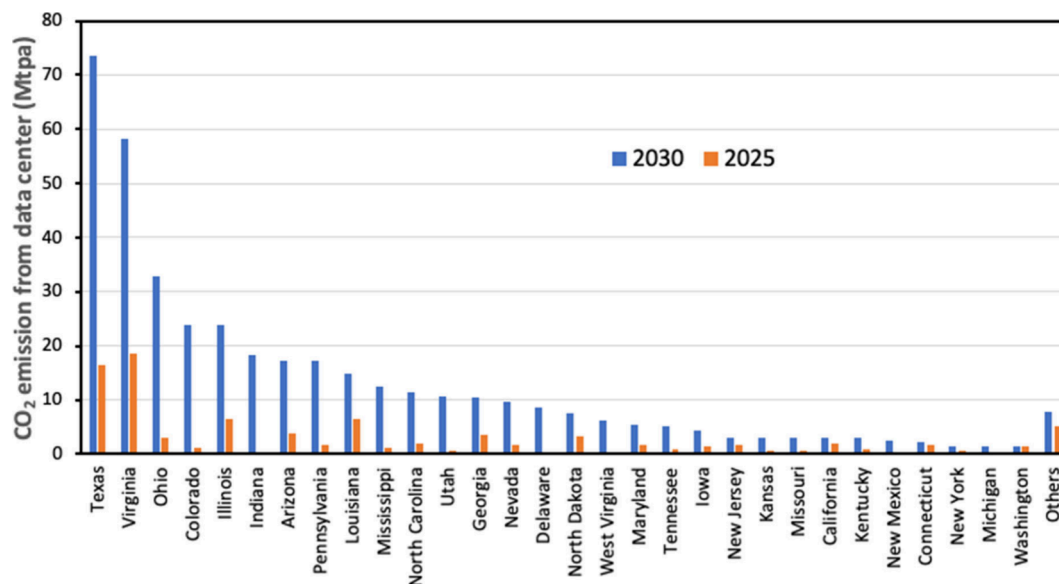


Figure 10. Estimated CO₂ emission from U.S. DCs in 2025 and 2030.



Figure 11. DC-emitted CO₂ from the lower 48 states of the U.S. in 2025.



Figure 12. Estimated DC-emitted CO₂ in the lower 48 states of the U.S. in 2030.

(Table 6). Arizona has no bordering states with saline aquifers. The states of Virginia, Missouri, Delaware, Iowa, New Jersey, Minnesota, and Wisconsin have the option to transport their DC-emitted CO₂ to a bordering state for sequestration. However, the length of the pipeline needed for such CO₂ transport (most over 200 miles) may be a barrier. In total, more than 90% of DC-related CO₂ emission can be mitigated by CCS, whether in-state or out-of-state.

3.4.3. CFPPs with CCS for Powering U.S. DCs. CFPPs are experiencing a rapid decline in the U.S., with capacity expected to fall significantly as owners pivot to cheaper natural gas and renewables.⁸¹ Approximately 12.3 GW of coal capacity was scheduled for retirement in 2025, a 65% increase from 2024, driven by high operating costs, aging infrastructure, and stricter environmental regulations. Coal's share of U.S. electricity generation has dropped from over 50% in the early 2000s to 16% in 2023 (Figure 8). The increasing

availability and lower cost of natural gas⁸² have made it the preferred source for base-load power over coal. Although CFPPs equipped with CCS can provide low-carbon electricity for U.S. DCs, it is unlikely that a significant number of new CFPPs will be built in the U.S. given the rapidly declining trend. Consequently, we expect natural gas with CCS to be the preferred option for powering new DCs.

3.4.4. NGCC with CCS for Powering U.S. DCs. There is an abundance of natural gas resources in the U.S. with production concentrated in Texas, Pennsylvania, Louisiana, West Virginia, Oklahoma, New Mexico, and Colorado. Furthermore, the U.S. has an extensive network of natural gas pipelines consisting of three million miles of main pipelines that link natural gas production areas to storage facilities and consumers.⁸³ This makes natural gas the primary energy source of electricity production in the U.S.⁸⁴

Table 4. U.S. Cities where DC-Emitted CO₂ Will Be over 5 Mtpa in 2030

	City	State	CO ₂ emission 2025 (Mtpa)	CO ₂ emission 2030 (Mtpa)
1	Columbus	Ohio	2.30	28.96
2	Delta	Colorado	0.00	22.79
3	Chicago	Illinois	6.56	22.42
4	Austin	Texas	5.47	17.90
5	Phoenix	Arizona	3.81	16.14
6	Panhandle	Texas	1.94	15.51
7	Shreveport	Louisiana	6.27	14.92
8	Ashburn	Virginia	9.79	14.82
9	Fredericksburg	Virginia	0.35	12.84
10	Dallas-Fort Worth	Texas	5.67	12.63
11	Mississippi	Mississippi	1.21	12.40
12	Abilene-Wichita Falls	Texas	1.55	11.24
13	Pittsburgh	Pennsylvania	0.99	10.06
14	Charlotte	North Carolina	1.73	10.04
15	Atlanta	Georgia	2.47	9.50
16	Reno	Nevada	1.24	9.31
17	Richmond	Virginia	1.20	8.47
18	South Bend	Indiana	0.12	8.41
19	Wilmington	Delaware	0.03	8.33
20	Manassas	Virginia	3.32	7.57
21	Delta	Utah	0.01	6.92
22	Shenandoah	Virginia	0.15	6.69
23	Allentown	Pennsylvania	0.46	5.24

Figure 15 shows the locations of oil and gas reservoirs in the lower 48 states. By comparing Figures 13 and 15, it can be seen that in Texas, Colorado, Louisiana, Mississippi, and Pennsylvania, the DC-related CO₂ clusters are relatively close to in-state natural gas reservoirs and saline aquifers. Therefore, it makes sense to locate the DCs close to both natural gas reservoirs and saline aquifers for the ease of energy supply and CCS. In such situations, NGCC with CCS (NGCC + CCS)

can be a viable solution for supplying low-carbon electricity to DCs.^{70–73}

We also estimate the natural gas needed to power all U.S. DCs in the 6 year period between 2025 and 2030. In our study, we use the typical heat rate⁸⁵ to estimate the amount of energy, expressed in British thermal unit (BTU), used by a power generator (a gas turbine or NGCC) to generate 1 net kWh of electricity with a given thermal efficiency.^{71,86} At 100% ideal thermal efficiency, 3412 BTU is required to generate 1 kWh. Therefore, at an average thermal efficiency of 45% in the gas industry in 2025, the heat rate is 7600 MM BTU/MWh. This means that a 1 MW gas power plant/turbine will consume 64.14 MM ft³ of natural gas per year. The amount of natural gas required to meet DC power demand (assuming 100% via a mature power generation technology such as a gas turbine) can thus be estimated. As the MW power demand grows over the years, additional gas turbines can be added accordingly. The choice of gas turbine as the power supply also enables the flexibility of installing gas turbines on site of a large DC campus or at a combined cycle gas plant owned by the state grid somewhere else by taking advantage of the stable and well-established natural gas pipeline network in the U.S.

Our calculations show that to power the entire DC industry, the amount of natural gas required was at least 2570 billion cubic feet (BCF) in 2025, representing 6.8% of 2024 dry gas production in the U.S. and 0.4% of the total proven gas reserves in the U.S.⁸⁷ This requirement will grow to at least 10 860 BCF in 2030, representing 28.8% of 2024 dry gas production and 1.8% of the total proven U.S. gas reserves. These numbers indicate that the natural gas industry is expected to ramp up more gas production in order to meet the power demands from more gas turbine installation, and there is ample proven gas reserves to power DCs since the U.S. gas reserves have a growth rate ranging between 11% in 2022 and 32% in 2021 because of discovery in Alaska.^{88–90}

Furthermore, the large and growing number of power purchase agreements (PPAs) signed and publicly disclosed by various large DC players, typically with a 20 year term,

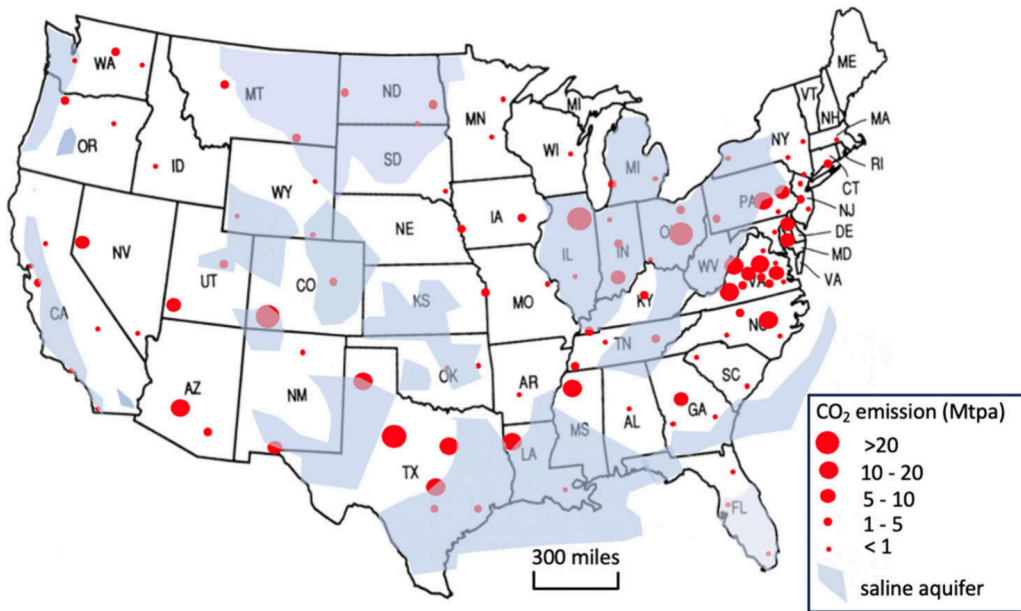
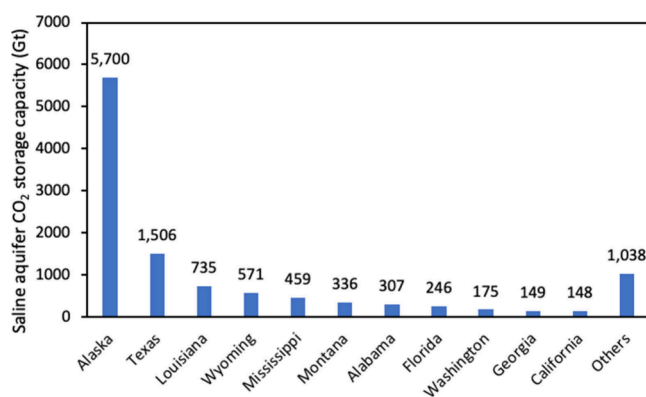


Figure 13. Locations of saline aquifers relative to CO₂ emission clusters in the lower 48 states of the U.S. in 2030 (source: authors).

Table 5. CO₂ Storage Capacity in Saline Aquifers and DC-emitted CO₂ in U.S. States

#	State	CO ₂ storage capacity in aquifer (Gt)	DC-emitted CO ₂ in 2025 (Mtpa)	DC-emitted CO ₂ in 2030 (Mtpa)	Years of CO ₂ injection
1	Alaska	5700.00	0.36	0.36	>100
2	Texas	1505.79	16.49	73.57	>100
3	Louisiana	734.55	6.33	14.98	>100
4	Wyoming	570.92	0.75	0.75	>100
5	Mississippi	459.15	1.21	12.40	>100
6	Montana	335.74	0.004	1.32	>100
7	Alabama	307.34	0.014	0.014	>100
8	Florida	246.45	0.79	0.79	>100
9	Washington	175.26	1.34	1.44	>100
10	Georgia	148.70	3.47	10.51	>100
11	California	147.55	2.03	2.90	>100
12	North Dakota	136.60	3.34	7.43	>100
13	Colorado	131.11	1.18	23.97	>100
14	New Mexico	129.29	0.14	2.46	>100
15	Utah	88.65	0.77	10.74	>100
16	Illinois	80.75	6.57	23.90	>100
17	Oklahoma	76.87	0.11	0.11	>100
18	Indiana	66.67	0.23	18.30	>100
19	Nebraska	54.47	0.09	0.42	>100
20	Kentucky	46.43	0.96	2.89	>100
21	Michigan	45.56	0.35	1.47	>100
22	Kansas	34.40	0.59	3.02	>100
23	Oregon	33.25	0.86	1.34	>100
24	South Carolina	31.07	0.43	0.43	>100
25	Arkansas	21.20	0.014	0.014	>100
26	Pennsylvania	17.34	1.74	17.29	>100
27	West Virginia	11.19	0	6.23	>100
28	Ohio	9.91	2.93	32.97	>100
29	South Dakota	7.04	0.03	0.49	>100
30	North Carolina	6.51	2.01	11.50	>100
31	New York	4.37	0.53	1.54	>100
32	Maryland	1.88	1.83	5.44	>100
33	Tennessee	1.85	1.03	5.16	>100
34	Idaho	0.15	0.028	0.028	>100
34	Virginia	0.86	18.68	58.21	<100
36	Arizona	0.42	3.89	17.39	<100
37	Missouri	0.10	0.59	3.02	<100
38	Delaware	0.04	0.03	8.53	<100
39	Nevada	0	1.65	9.72	0
40	Iowa	0	1.43	4.30	0
41	New Jersey	0	1.63	3.08	0
42	Connecticut	0	1.65	2.12	0
43	Minnesota	0	0.61	0.66	0
44	Massachusetts	0	0.49	0.56	0
45	Wisconsin	0	0.42	0.42	0
46	Hawaii	0	0.05	0.05	0
47	Maine	0	0.006	0.006	0
48	New Hampshire	0	0.002	0.002	0
49	Rhode Island	0	0	0	NA
50	Vermont	0	0	0	NA
Total		11369	90	404	

indicates a strong interest in long-term sustainable power sources and a healthy demand growth rate of third-party power suppliers, not to mention the large number of undisclosed

**Figure 14.** Medium CO₂ storage capacity in saline aquifers in various U.S. states (data source: DOE).⁶¹**Table 6. Out-of-State CO₂ Sequestration Options for States with No In-State Sequestration Option**

State	DC-emitted CO ₂ in 2025 (Mtpa)	DC-emitted CO ₂ in 2030 (Mtpa)	CO ₂ injection to neighboring state	Length of CO ₂ pipeline needed (miles)
Arizona	3.89	17.39	Not possible	Not applicable
Virginia	18.68	58.21	WV, PA	240/272
Missouri	0.59	3.02	IL, KS, OK	263/520/333
Delaware	0.03	8.53	WV, PA	278/246
Iowa	1.43	4.30	IL	305
New Jersey	1.63	3.08	PA	184
Minnesota	0.61	0.66	NS, SD	242/348
Wisconsin	0.42	0.42	IL	297
Total ^a	23.39	78.22		

^aExcluding Arizona.

PPAs in the making due to ongoing contractual negotiations.^{91–93}

3.4.5. Other Options to Decarbonize U.S. DCs. At present, natural gas is the dominant fuel source for electricity generation in the U.S. with nuclear and coal in the second and third place, respectively (Figure 8). Renewable energies (wind, solar, hydropower, geothermal, and biomass) contributed to only 22.1% of U.S. electricity generation in 2023.

Of the five major renewable energies, hydropower and geothermal are location specific and not widely available throughout the U.S. Wind energy is concentrated in the Central Plains (Texas, Oklahoma, and Iowa), the Midwest, and California, which have relatively flat terrain with high wind speed essential for utility-scale wind turbines. Solar PV is mostly concentrated in the “Sunbelt region” consisting of the Southwest, California, and Texas, which has the highest solar resources. Unfortunately, neither solar PV nor wind is available continuously, and thus they cannot be used as a base-load even coupled with expensive battery storage. In fact, the EIA classifies wind, solar PV, hydroelectricity, and solar PV-battery hybrid as resource-constrained technologies.⁹⁴ Of the renewables, only biomass and geothermal are classified as dispatchable technologies. However, biomass is too expensive to be used on a utility scale, and geothermal is limited to certain locations in the U.S. with geothermal resources. In most cases, the base-load has to come from either fossil fuels or nuclear energy.

Nuclear energy is being considered as a power source for U.S. DCs. Unfortunately, the long permitting process means

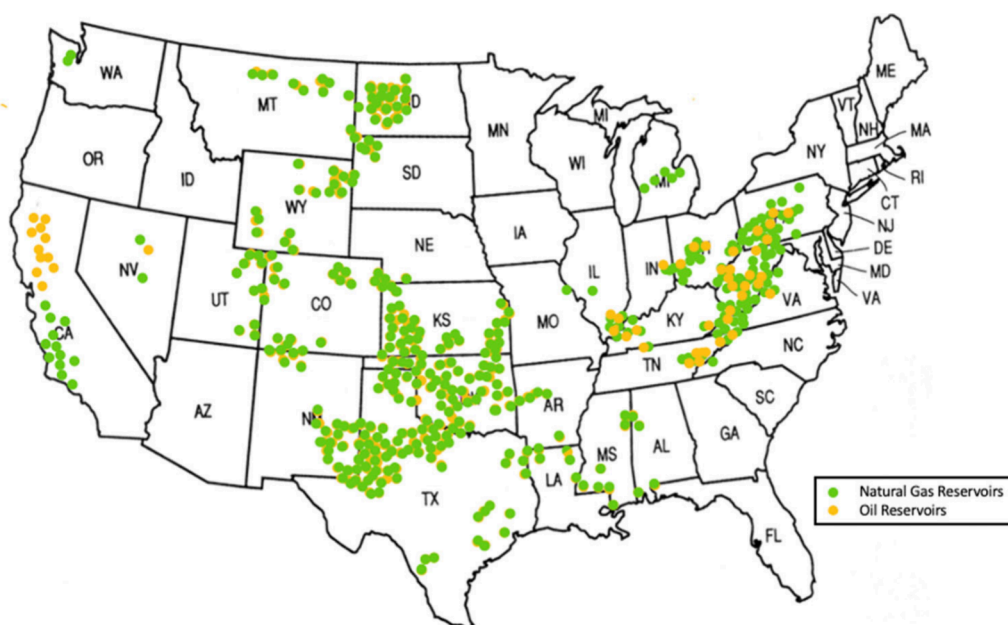


Figure 15. Location of onshore oil and gas reservoirs in the lower 48 states of the U.S.

that the time from planning to completion of a nuclear power plant often takes close to a decade.^{68,95,96} Even SMRs can take a significant amount of time for permitting, and the first commercial SMR probably will not be available before 2030.^{69,97,98}

CFPPs equipped with CCS can be used as an energy source for U.S. DCs. However, compared to natural gas, the combustion of coal produces approximately twice the amount of CO₂ for the same unit of energy produced.^{99,100} In addition, there are more GFPPs than CFPPs in the U.S. Consequently, NGCC + CCS is the preferable option for DCs.

3.4.6. Comparison of Levelized Cost of Electricity (LCOE) of New Power Plants. In their *Annual Energy Outlook 2025*, the EIA gives a comparison of new built power plants, as shown in Table 7.⁹⁴ The reported LCOE includes

Table 7. Estimated LCOE for New Power Plants Entering Service in 2030 (Source: Data from EIA (2025)⁹⁴ CC BY 4.0)

Average LCOE (\$2024/MWh)	
Dispatchable Technologies	
Advanced nuclear (including SMR)	81.45
Biomass	80.85
Natural gas combined cycle (NGCC)	64.55
NGCC + CCS	48.78
Resource-Constrained Technologies	
Offshore wind	88.18
Hydroelectricity	58.54
PV-battery hybrid	53.44
Solar PV	31.86
Onshore wind	29.58

levelized capital cost, operation and maintenance cost, transmission cost, tax credit, and carbon capture tax credit. The levelized CCS cost includes the negative cost of captured carbon credit and the cost of transporting CO₂ to saline aquifers for sequestration. The Section 45Q Internal Revenue Service (IRS) tax credits^{101,102} under the Inflation Reduction

Act (IRA) of 2022 are available for the first 12 years of a plant's operation if construction starts before 2033.

It can be seen that of the dispatchable technologies entering into service in 2030, NGCC + CCS has the lowest average LCOE. Only solar PV and onshore wind have lower LCOEs. However, they are resource-constrained or nondispatchable technologies.

The conclusion of this EIA study agrees with that of another recent study showing that NGCC + CCS has a lower LCOE compared to a newly built nuclear power plant.¹⁰³

4. LIMITATIONS OF STUDY

This study is based on several important assumptions. First, within our study period between 2025 and 2030, we consider only DCs where their power requirement is announced. DCs that are planned but with their power requirement unannounced due to competitive or other reasons are not included, since no data are available for consideration. Second, if a company does not specify the source of their power, we assume the power is taken from the state grid, and we calculate the CO₂ emission based on the energy mix of the state grid, which is public information. We further assume that the energy mix of the state grid will remain unchanged between 2025 and 2030 to calculate the DC-emitted CO₂ in 2030. Some companies announce their PPAs, and we use them to calculate the DC-emitted CO₂. On the basis of these limitations, we consider our estimates of DC-emitted CO₂ to be conservative, yet informative, for scientists, engineers, and policymakers interested in DC decarbonization.

Another limitation of this study is that we assume that CO₂ is stored in saline aquifers rather than oil or gas reservoirs. This assumption is justified because according to studies by the DOE, oil and gas reservoirs constitute only 2.4% of CO₂ storage capacity in North America, compared to 96.7% by saline aquifers.⁶¹ However, there are nine states with CO₂ storage in oil and gas reservoirs larger than 1 Gt: Texas, West Virginia, Louisiana, New Mexico, California, Oklahoma, Utah, Colorado, and Pennsylvania. These states have an additional option to store CO₂ in depleted oil or gas reservoirs. However,

the suitability of CO₂ injection into oil and gas reservoirs depends on the state of depletion and whether the CO₂ is miscible with the oil at reservoir conditions. In principle, CO₂-enhanced oil recovery (EOR) can be more economical than CO₂ sequestration in saline aquifers, due to the value of incremental oil produced. However, implementation of CO₂-EOR requires detailed laboratory and reservoir simulation studies, which are outside the scope of this study. Gas reservoirs can store CO₂ if they are depleted. Before depletion, injection of CO₂ into a gas reservoir may introduce CO₂ in the produced gas, which must be stripped before it can be sold. In these nine states, a change in the CO₂ sink from saline aquifers to oil and gas reservoirs will not affect the conclusions.

5. DISCUSSION

The impact of DC-emitted CO₂ on the total CO₂ emission in the U.S. is shown in Figure 16. This figure plots the historical



Figure 16. Historical and projected CO₂ emission in the U.S. (adapted from Lau and Tsai¹⁰⁴ CC BY).

and projected CO₂ emission in the U.S.¹⁰⁴ In this figure, we have indicated the expected increase in CO₂ emission from U.S. DCs in 2030 from this study. It can be seen that the incremental CO₂ emission from DCs will significantly affect the rate of decline of CO₂ with time in the U.S. Figure 16 shows that CO₂ emission in the U.S. has been declining since 2002, and if the current trend continues, CO₂ emission will decline to about 4.5 Gtpa in 2030. This shows that the U.S. has exited from being a “polluter”, where CO₂ emission is increasing with time, to being a “decarbonizer”, where CO₂ emission is decreasing with time.¹⁰⁴ However, additional CO₂ emission from DCs will be 0.4 Gtpa by 2030, according to this study. This means that CO₂ emission will be at about 4.9 Gtpa by 2030, which will put it at the same level as that in 2023, thus mitigating seven years of CO₂ decline assuming no CO₂ emission from DCs. This is quite significant. DC-emitted CO₂ will significantly reduce the overall rate of decline of CO₂ emission in the U.S. unless this incremental CO₂ emission is mitigated by CCS.

Our study shows that CCS can be used to mitigate CO₂ emitted from either CFPPs or GFPPs used to power DCs. OFPPs play an insignificant role in powering DCs since they play a significant role in electricity generation only in the states of Alaska and Hawaii, which do not have many DCs. Furthermore, our analysis shows that 18 states will have DC power capacity exceeding 10% of the total state power capacity by 2030 and will likely need to add new capacity (Figure 5).

Results in this study also suggest that natural gas with CCS will be a viable option for new power capacity for future DCs.

6. CONCLUSIONS

Analysis of data in the public domain shows that the power capacity of U.S. DCs will grow from 40 GW in 2025 to 169 GW in 2030. The states with the highest DC growth rate will be Texas, Virginia, Pennsylvania, Ohio, Arizona, Colorado, Utah, and Illinois. Using the announced power source of future DCs and the energy mix of the state power grids, this study estimates that DC-emitted CO₂ will increase from 90 Mtpa in 2025 to 404 Mtpa in 2030. A CO₂ source-sink mapping exercise has shown that 34 states have enough capacity in saline aquifers to store more than 100 years of DC-emitted CO₂ beyond 2030. Of the 16 states that do not have adequate CO₂ storage, six have the option of sequestering CO₂ in a bordering state. These results suggest that fossil fuel power plants equipped with CCS are a viable option to decarbonize U.S. DCs. Furthermore, NGCC + CCS is the preferred option due to the abundance of gas reserves, the proximity of gas fields to saline aquifers, and the lower CO₂ emission of GFPPs compared to CFPPs.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.energyfuels.6c01309>.

Share of state electricity generation by fuel type and conversion factors used to calculate CO₂ emission from plant capacity (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Hon Chung Lau – Low Carbon Energies LLC, Bellaire, Texas 77401, United States; Department of Chemical and Biomolecular Science, Rice University, Houston, Texas 77005, United States; orcid.org/0000-0001-7749-1256; Email: hclau@lowcarbonenergies.com

Author

Steve C. Tsai – Low Carbon Energies LLC, Bellaire, Texas 77401, United States

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acs.energyfuels.6c01309>

Notes

The authors declare no competing financial interest.

■ NOMENCLATURE

AI	Artificial intelligence
BTU	British thermal unit
CAGR	Compound annual growth rate
CCS	Carbon capture and storage
CFPP	Coal-fired power plant
DC	Data center
DOE	Department of Energy
EIA	Energy Information Agency
EOR	Enhanced oil recovery
GFPP	Gas-fired power plant
IEA	International Energy Agency
IRA	Inflation Reduction Act

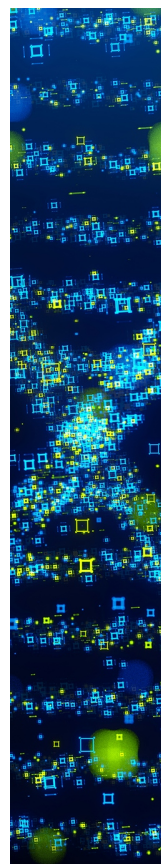
IRS	Internal Revenue Service
LCOE	Levelized cost of electricity
NGCC	Natural gas combined cycle
NGCC + CCS	Natural gas combined cycle with carbon capture and storage
OFPP	Oil-fired power plant
PPA	Power purchase agreement
PV	Photovoltaic
IT	Information technology
SMR	Small modular reactor
BCF	10 ⁹ standard cubic feet
GW	10 ⁹ W
Gtpa	10 ⁹ metric tons per annum
MW	10 ⁶ W
Mtpa	10 ⁶ metric tons per annum
m _{CO₂}	Mass of CO ₂ stored, kg
ρ _{CO₂}	Density of CO ₂ at reservoir conditions, kg/m ³
A	Aquifer area, m ²
h	Average sand thickness, m
φ	Average porosity of sand, fraction
E	CO ₂ efficiency factor, fraction

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